Robotics Information Modelling (RoIM): A Framework for Robot-Ready Building Information

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> **Disclaimer: framework under development, not ready for implementation.** RoIM is a draft proposal for discussion and research. The definitions, RoIM levels (R0 to R4), property set, IDS rules, OpenUSD schema and workflows in this paper have not been validated on any live project or with any robot. Do not use them to specify, design, check, accept or operate construction work or robots, or to make safety-related decisions. RoIM does not certify robot safety, model accuracy or compliance with any law, contract or standard. Content will change between versions. RoIM is an independent initiative and is not endorsed by buildingSMART International, the Alliance for OpenUSD, NVIDIA or any other organisation named here.

Companion repository (draft specification files and research tools): github.com/ernixdotio/roim-framework. Cite all versions as doi:10.5281/zenodo.24099902.

Abstract

Robotics Information Modelling (RoIM) is the practice of structuring building information so that machines, not only people, can act on it safely. It extends Building Information Modelling (BIM) with what construction robots need: survey-grade geometry, machine-readable semantics, the state of the site over time, and a feedback loop from robot to model.

Research has connected BIM to robots for over a decade, but each team builds its own bridge. There is no shared name, no graded level of readiness and no common property set. This paper proposes all three: a definition of RoIM, five RoIM Levels of Information (R0 to R4), and a draft IFC property set, `Pset_RobotReadiness`. The aim is a common language that BIM coordinators, robotics engineers and contractors can specify, check and deliver against.

1. Introduction

Building models are written for human readers, and robots cannot use them reliably. A coordinator reads a drawing and fills the gaps from experience. A robot cannot. It needs exact positions, tolerances, surface properties, access constraints and the current state

of the work, and most production BIM supplies none of these in a form a machine can trust.

The industry is moving anyway. Layout robots set out on slabs, quadrupeds scan sites weekly, drones capture progress, and drywall, drilling and printing robots are leaving pilot projects. Each one consumes building information, and each integrator currently rebuilds that information by hand for every project.

Research confirms the gap. Studies report that 4D BIM lacks the element-level information a robot needs to plan within the construction schedule, that current modelling practice leaves too little detail for robotic construction, and that BIM is seldom used for robot navigation because models are unreliable and do not match localisation algorithms. One review describes the work as building a bridge between two communities that do not yet share a language.

RoIM is that shared language. It treats robot readiness as a deliverable that can be specified in an Exchange Information Requirement, modelled, checked and accepted, just as LOD and LOIN are today.

2. Definition and scope

Robotics Information Modelling (RoIM) is the structured creation, management and exchange of building information to the geometric accuracy, semantic depth and currency that an autonomous or semi-autonomous machine needs to localise, navigate, set out, build, inspect or verify work in the built environment.

A **robot-ready model** is a building information model that meets a stated RoIM level for a stated robotic task.

The abbreviation is **RoIM**. It avoids "RIM", which is already used for Railway Information Modelling and Road Information Models.

In scope

- Construction-phase robots: layout and setting out, reality capture and progress tracking, deviation analysis, drilling and fixing, drywall and finishing, printing and assembly.
- Operations-phase robots: inspection, cleaning, security and facility-management platforms working in a finished building.
- Aerial and ground platforms, wheeled, legged and tracked.

Out of scope

- Describing robot hardware itself. That is covered by robot information models such as HRIM and IEEE 1872.

- Off-site factory automation, where the CAD/CAM chain already supplies machine data. RoIM can still feed it.

RoIM sits on top of existing standards: IFC (ISO 16739), ISO 19650 information management and Level of Information Need (EN 17412-1). It adds a robot-facing layer and invents no new schema where an existing one will do.

3. Related work

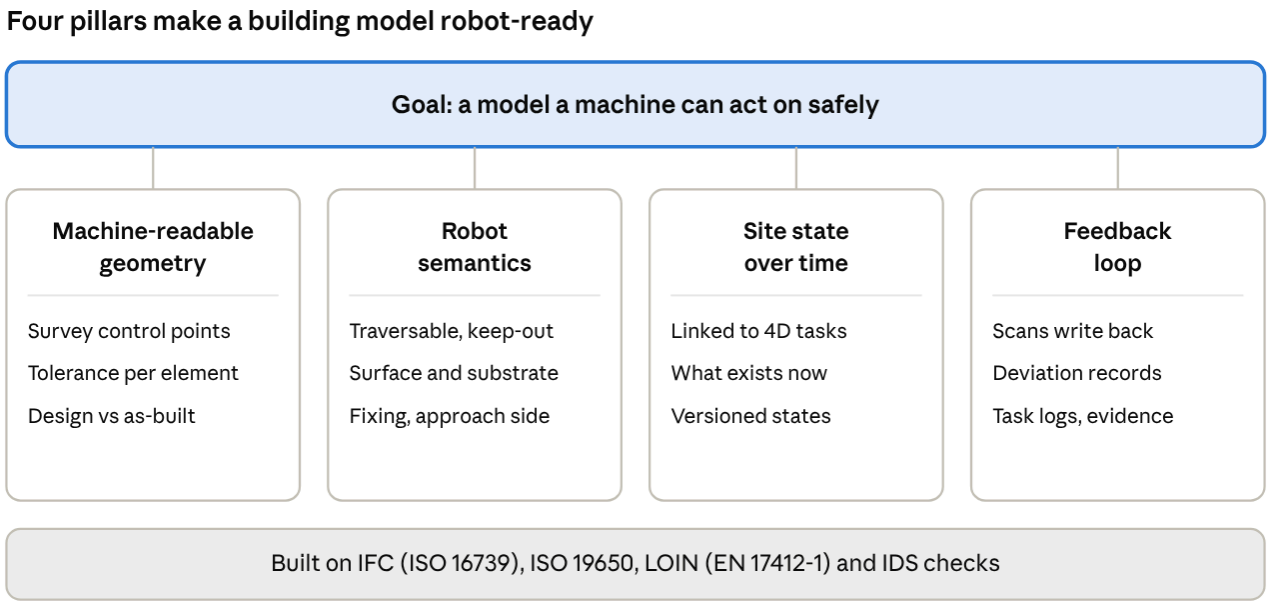
A lot of work already connects BIM and robots, but none of it names a discipline or grades readiness. RoIM builds on the work below and generalises it.

Work	What it does	What RoIM adds
BIM-robotics systematic reviews (Byers et al., 2022; LSU, 2025)	Maps how BIM supports robot navigation, task planning and progress tracking	A single framework for those uses
Information Modeling for 4D BIM-Based Robot Task Planning (ASCE CRC, 2024)	Identifies the 4D BIM information a drywall robot needs	Generalises this to every task through levels
BIM to Robotic Construction information exchange (NSF)	Maps BIM exchanges to mobile-robot system elements and raises LOD with Dynamo	A standard property set instead of project-specific scripts
BIM2RDT (Akhavian et al., 2025)	Turns static BIM into robot-ready site digital twins using quadruped scans, IoT and LLM reasoning	A vendor-neutral definition of "robot-ready" that such systems can target
BIM-based robot navigation (Pauwels et al.)	Shows BIM can drive robot navigation through RDF and JSON; calls for modelling guidelines and a robot world model	Those modelling guidelines
buildingSMART BIM-ROS Integration use case	An approved example use case linking IFC and ROS	A full set of RoIM use cases to submit beside it
HumanTech bSDD dictionary	Robotics and safety terms in the buildingSMART Data Dictionary	Graded readiness on top of those terms

Work	What it does	What RoIM adds
Robot-Oriented Design (Bock; Correa, 2016)	Design buildings so robots can produce them	Covers the information, not only the design
HRIM (2018)	An information model for robot hardware modules	Out of scope: RoIM models the building, not the robot

4. The four pillars of RoIM

A model is robot-ready when it carries four kinds of information on top of normal BIM.



RoIM · four pillars on an openBIM foundation

- **Machine-readable geometry.** Positions a machine can trust to a stated tolerance, tied to survey control. Design intent and as-built geometry are kept apart.
- **Robot semantics.** Properties a human infers but a robot must be told: where it can drive, where it must not go, what surface it touches, what sits behind it and how an element is fixed.
- **Site state over time.** What exists on site now, linked to the 4D programme, so a robot does not plan against walls not yet built.
- **Feedback loop.** Results from robots, such as scans, deviations and task logs, are written back into the model. Without this the model goes stale after the first mission.

5. RoIM Levels of Information (R0 to R4)

A RoIM level states what a machine can safely do with a model. Levels are set per element and per task, as LOIN is, not for the whole model. One project can need R3 for slab set-out and R1 for façade inspection.

Level	Name	A robot can…	Minimum information	Example task
R0	Visual	Nothing autonomous; humans interpret the model	Standard design BIM, any LOD	Design review
R1	Navigable	Localise and move through the space	Georeferenced model, survey control points, floors, walls, openings and stairs with traversable or blocked status, keep-out zones	Autonomous progress scan
R2	Verifiable	Compare as-built against design and report deviation	R1 + tolerance per element, IFC class and GUID kept stable, as-designed state flagged, scan-to-model registration targets	Weekly deviation analysis
R3	Actionable	Set out or work on an element	R2 + survey-grade setting-out points, fixing method, surface and substrate data, approach side and reach envelope, task sequence linked to 4D	Layout printing, drilling, drywall
R4	Closed-loop	Write results back and keep the model current	R3 + as-built state per element, robot task log, deviation records and evidence links written to the model, version control on state	Robot-maintained digital twin

Each level is checkable. A model either carries the required properties with valid values or it does not, so RoIM compliance can be tested automatically with IDS (Information Delivery Specification) rules.

6. Draft property set: Pset_RobotReadiness

RoIM can be delivered in today's tools with one custom property set. It works in Revit as shared parameters exported to IFC, and in any IFC authoring tool. Version 0.1 is proposed below for public comment. Prefixes follow buildingSMART guidance for non-standard sets.

Property	Type	Values or unit	From level
RoIM_Level	Enumeration	R0, R1, R2, R3, R4	R0
RoIM_TaskScope	Label list	e.g. Navigate, Scan, SetOut, Drill, Finish, Inspect	R1
Traversable	Boolean	True if a ground robot may cross or occupy it	R1
KeepOutZone	Boolean	True if no robot may enter it	R1
ControlPointRef	Label	ID of the survey control point the element is set out from	R1
PositionTolerance	Length	± mm, design tolerance for placement	R2
AsDesignedState	Enumeration	Design, Approved for construction, Superseded	R2
RegistrationTarget	Boolean	True if usable as a scan-registration feature	R2
SettingOutPoints	Label	Reference to a point set (CSV or IfcGrid/IfcAnnotation)	R3
SurfaceType	Label	e.g. Concrete fair-faced, Gypsum board, Glazing	R3
SubstrateMaterial	Label	Material behind the surface, for drilling and fixing	R3
FixingMethod	Label	e.g. Anchor M10, Screw, Adhesive	R3
ApproachSide	Enumeration	Front, Back, Top, Underside, Any	R3
ScheduleTaskRef	Label	4D task ID that the element belongs to	R3
AsBuiltState	Enumeration	Not started, In progress, Installed, Verified, Non-conforming	R4
MeasuredDeviation	Length	mm, worst deviation from last verification	R4

Property	Type	Values or unit	From level
LastVerifiedDate	Date	ISO 8601	R4
VerificationSource	Label	Robot or scan ID plus a link to the evidence	R4

The set is meant to be released as an IDS file and a Revit shared-parameter file. Teams can then require it in an EIR and check delivery automatically. Once stable, it should be proposed for the buildingSMART Data Dictionary.

7. Reference workflow

RoIM fits into an ISO 19650 project as one more information requirement and one more loop. A typical implementation:

1. **Specify.** The appointing party states, in the EIR, the RoIM level needed per element group and robotic task (e.g. R3 for slab set-out on levels 2 to 10).
2. **Model.** Authors add `Pset_RobotReadiness` properties as shared parameters. Survey control and setting-out points are linked to the project base point and georeferencing.
3. **Check.** The model is tested against the RoIM IDS before it is issued. Failures go back to the author like clash results.
4. **Export.** IFC is exported and parsed (e.g. with IfcOpenShell) into what the robot stack uses: occupancy maps, navigation graphs, setting-out point files, and labelled geometry for simulation.
5. **Simulate.** The task is rehearsed in a simulator such as NVIDIA Isaac Sim or Gazebo on the actual model. IFC classes give free semantic labels for synthetic training data.
6. **Execute.** The robot localises with SLAM (e.g. FAST-LIO2 on ROS 2), registers to the model's targets and does the task.
7. **Feed back.** Scan and task results are compared with the model, and deviation and as-built state are written back to the R4 properties. The updated model drives the next mission.

Steps 6 and 7 repeat for every mission. That loop turns a static design model into a model the site keeps current.

8. A new role: the RoIM coordinator

Robot readiness needs an owner, and the natural owner is the BIM coordinator. The RoIM coordinator:

- sets RoIM levels per task with the contractor and robotics provider
- writes the RoIM clauses in the EIR and BIM Execution Plan
- runs IDS checks and holds models to the agreed level before release to robots
- owns the survey-control link between model and site
- manages the feedback loop and the as-built state in the model

This turns BIM coordination from a drawing-production role into the controller of the data that machines rely on on site.

9. Limitations and future work

This is a position paper, version 1.0. The levels and property set are proposals that have not yet been validated on a live project.

- **Validation.** Levels R1 to R4 need testing on real projects with real robots, starting with set-out and deviation analysis.
- **Tolerances.** Default tolerances per element and task should come from field data, not opinion.
- **Dynamic state.** IFC is a snapshot format. R4 may need a live layer (e.g. a graph database or digital-twin platform) synced to IFC, not IFC alone.
- **Safety.** RoIM does not certify robot safety. Interaction with ISO 10218 and site safety rules needs its own work.
- **Standardisation.** The property set should move to the buildingSMART Data Dictionary, and RoIM use cases should be submitted to the buildingSMART use-case platform.

10. Conclusion and call for collaboration

Robots are arriving on construction sites faster than the information they need. RoIM gives the industry a name for that information, five levels to specify it, and a property set to deliver it with today's tools.

The next step is a pilot: one building, one robot task, modelled to R2 or R3 and measured. BIM managers, contractors, robotics companies and researchers who want to test, critique or extend RoIM are invited to get in touch. The property set and IDS will be published openly so anyone can use and improve them.

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